RED RIVER COLLEGE POLYTECHNIC · WORKSHOP HANDOUTS

Digital World Building

Scoping worksheets: Consider → Scope → Build

Work through these in your team of three. Each worksheet feeds the next. By the end you will have a scoping package that a person, or an AI coding agent, can build from.

#	Worksheet	Use it during
01	World Seed: The Ten Considerations	Consider
02	Identity, Ownership & Ethics Check	Consider
03	World Brief	Scope
04	AI Co-Authoring Log: the 4 Ds	Scope
05	Spatial Plan	Scope
06	Asset List & Budgets	Scope / Tools
07	Interactions & Build Plan	Build
08	Choose Your Tools	Tools
09	Playtest Notes	Playtest

Team / names	
World name	

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WORKSHEET 01

World Seed: The Ten Considerations

Draw a seed card (or invent your own). Answer each question in one line. Rough is fine. You are finding the decisions, not making them perfect.

Team / names		World name	

Our seed	

Consideration	Question	Our one-line answer
Purpose & audience	Why does this world exist, and who is it for?	
Platform & constraints	Browser, phone, headset, PC? Seated or standing?	
Experience & verbs	What does a visitor DO in the first 60 seconds?	
Space & wayfinding	How big? What zones? What is the landmark?	
Story & lore	What happened here before the visitor arrived?	
Rules & systems	What is true here: time, weather, physics, economy?	
Look, sound & mood	Three mood words. One colour. One sound.	
Comfort & access	How do we avoid motion sickness and exclude no one?	
Culture & ethics	Whose places, symbols or stories appear? Who said yes?	
Scope & feasibility	What can we really build? What will we NOT build?	

Reflect

Which question was hardest to answer? That is where your world needs the most thinking.

WORKSHEET 02

Identity, Ownership & Ethics Check

Answer these before you build. They are expensive to fix later, and they shape who feels welcome in your world.

Team / names		World name	

Identity

Question	Our answer
How do visitors appear? (avatar, no body, hands only, a scanned likeness?)	
Can people choose or change how they look? Any limits?	
What does the world remember about a visitor between visits?	
How do people stay safe around others? (mute, block, personal space, reporting)	

Ownership & data

Question	Our answer
Who owns what visitors make or bring into the world?	
What data do we collect (movement, voice, gaze, choices)? Why do we need each one?	
Can people leave with what is theirs? How?	

Culture & consent

Question	Our answer
Whose culture, place, people or likeness appear in our world?	
Who have we asked, and who said yes? Who else should we consult?	
Is any knowledge in our world held by a community? Who controls how it is shared? (See OCAP® principles.)	

OCAP® (Ownership, Control, Access, Possession) is a registered trademark of the First Nations Information Governance Centre (FNIGC).

WORKSHEET 03

World Brief

One page that every later decision answers to. Fill it by hand first. Mark anything you are guessing with [?] and list it under Open Questions.

Team / names		World name	

One-line pitch	A _____ where you _____ in order to _____.
Purpose	
Audience	
Platform & input	

Design pillars: the three rules every decision must serve

1	
2	
3	

Core experience

First 60 seconds	
Core loop	
Payoff / ending	

Zones (3 to 5)

ID	Zone name	Purpose	Landmark	Mood
Z1				
Z2				
Z3				
Z4				
Z5				

Key interactions (5 or fewer)

ID	Trigger (the visitor does…)	Response (the world does…)
I1		
I2		
I3		
I4		

Story, look and sound

Lore in 3 sentences	
Mood words (3)	
Palette (5 colours)	
Sound	

Comfort, culture and constraints

Comfort & access	
Culture & ethics	
Time / team / budget	
Performance target	fps: _____ triangles: _____ draw calls: _____
Out of scope	We are NOT building:
Open questions	

WORKSHEET 04

AI Co-Authoring Log: the 4 Ds

Use AI to expand and stress-test your brief, not to replace your judgement. Log what you asked, what came back, and what you kept.

Team / names		World name	

Direction: decide before you prompt

Our goal for using AI today	
What success looks like	
What AI must NOT decide for us	

Description: the prompt chain

#	Prompt (from the Scoping Toolkit)	Paste or summarise what you sent
1	Interview me: 8 questions, one at a time	
2	Draft the brief; mark guesses [ASSUMPTION]	
3	Critique it as a senior world designer	
4	Spatial plan + ASCII map, 1 unit = 1 m	
5	Asset list with triangle budgets	
6	Build spec for an AI coding agent	

Discernment: keep, change or cut

What AI suggested	Keep / change / cut	Why

What AI suggested	Keep / change / cut	Why

Discernment checklist

- ☐ Does it serve our three pillars, or does it just sound cool?
- ☐ Is anything stated as fact about a real place, culture or people that we have not verified?
- ☐ Is this the most generic version of the idea? What is the specific one?
- ☐ Can we actually build it in the time we have?
- ☐ Does it work for someone seated, new to games, or with low vision?
- ☐ Are all [ASSUMPTION] lines still visible until a human confirms them?

Delegation

AI drafts / builds	Humans own

WORKSHEET 05

Spatial Plan

Sketch your world from above. One square = _____ metres. Mark north, the main path, every zone, and the landmark. Draw arrows for important sightlines.

Team / names		World name	

Legend: ━ path ▒ edge / boundary ◯ zone ● node (where paths meet) ▲ landmark → sightline ★ spawn point

Zones and sizes

ID	Zone	Approx. size (m)	Connects to	Can you see the landmark from here?
Z1				
Z2				
Z3				
Z4				
Z5				
Z6				

Wayfinding check (Kevin Lynch's five elements)

Element	In our world, this is…
Paths: routes people move along	
Edges: boundaries (shoreline, wall, treeline)	
Districts: areas with their own character	
Nodes: meeting points, intersections	
Landmarks: things you navigate by	

Scale check

- ☐ 1 unit = 1 metre. Eye height 1.6 m, door 2.1 m, ceiling about 2.7 m.
- ☐ A first-time visitor can tell where to go within 10 seconds of arriving.
- ☐ Every zone can be seen from at least one other zone.
- ☐ Boundaries are natural (water, trees, cliffs), with invisible walls only as a last resort.

WORKSHEET 06

Asset List & Budgets

List every object you need. Decide its priority and where it comes from before anyone starts modelling.

Team / names		World name	

Budgets for the whole scene

Target device	Frame rate	Triangles	Draw calls	Texture memory

Starting points: desktop browser 60 fps; standalone headset (e.g. Quest) 72 to 90 fps with roughly 100k to 300k triangles and about 100 draw calls. Measure on your real device.

Asset list

Asset	Zone	Priority M / S / C	Source	Tri budget	Licence / credit

Priority: M = must have, S = should have, C = could have. **Source:** Build (Blender) · Blender + Claude (MCP) · ComfyUI (image → 3D) · CC0 library · Buy · Commission

Before you call an asset done

- ☐ Real-world scale, pivot at the base, facing +Z (or your engine's forward).
- ☐ Named consistently, e.g. zone_object_variant (dock_lantern_a).
- ☐ Under its triangle budget; textures power-of-two (512, 1024, 2048).
- ☐ Exported as GLB / FBX and tested in the engine, not just in Blender.
- ☐ Licence and source written in the ledger. AI-generated assets labelled as such.

WORKSHEET 07

Interactions & Build Plan

Every interaction needs a trigger, a response and feedback the visitor can notice. Then plan the build in stages, greybox first.

Team / names		World name	

Interaction spec

ID	Trigger	Response	Feedback (see / hear / feel)	Works with mouse / touch / VR?
I1				
I2				
I3				
I4				
I5				

Build stages

Stage	Goal	Done when…	Owner	Date
1 Greybox	Prove layout, scale and flow with boxes			
2 Blockout	Zones, paths, landmark, sightlines			
3 Art pass	Palette, style, lighting from the brief			
4 Life	Interactions, audio, story beats			
5 Polish	Effects, tuning, under budget			

Risks

What could stop us?	What we will do about it

WORKSHEET 08

Choose Your Tools

Pick the engine and asset pipeline that fit your brief, your team and your timeline. There is no best engine, only the best fit.

Team / names		World name	

Engines at a glance (fall 2026: check current terms before you commit)

	Godot	Unity	Unreal Engine	Web (A-Frame, Three.js)
Cost	Free, open source (MIT). No royalties.	Free Personal plan under $200K revenue or funding; paid seats above.	Free until a title earns $1M lifetime gross; then 5% royalty.	Free, open source.
Language	GDScript (Python-like), C#	C#	Blueprints (visual), C++	HTML + JavaScript
Strengths	Tiny, fast editor; gentle learning curve; good web export	Biggest tutorial library; widest XR and mobile support	High-end visuals; virtual production; Blueprints for non-coders	Runs from a link; no install; WebXR in the browser
Watch out for	XR support still maturing	Licence terms have changed before	Heavy hardware; limited web export	Performance ceiling; you build more yourself
Good for	Learning, stylised 3D and 2D, small teams	XR training, mobile, cross-platform	Photoreal worlds, film, high-end VR	Prototypes, spatial web, quick sharing

Which fits us? Tick what is true

- ☐ We need it to run in a browser from a link → Web, or Godot web export
- ☐ We need it on a standalone VR headset → Unity (most mature), Unreal, or WebXR
- ☐ Photoreal is a pillar → Unreal
- ☐ No one on the team codes yet → Unreal Blueprints, or Godot's GDScript
- ☐ We have weeks, not months → Web or Godot

Our engine	
Why	

Asset pipeline: where our 3D, images and video come from

Tool	What it is good for	We will use it for
Blender	Free 3D modelling, UVs, materials, export to GLB/FBX	

Tool	What it is good for	We will use it for
Blender + Claude (MCP)	Describe it in words; Claude writes and runs Blender Python to build, fix, batch-rename and export	
ComfyUI	Node-based, local generative AI: concept images, textures, video, image-to-3D meshes	
CC0 libraries	Poly Haven, Kenney, Quaternius: free models, textures and HDRIs	

AI asset guardrails

- ☐ Check the licence of every AI model, not only the output.
- ☐ Do not generate real people, logos or cultural designs without permission.
- ☐ Label AI-generated assets in the licence ledger.
- ☐ Clean up AI meshes before use: scale, triangle count, holes, UVs.

WORKSHEET 09

Playtest Notes

Watch someone who did not build it. Do not explain anything. Write down what they do, not what you hoped they would do.

Team / names		World name	

World tested	
Tester	
Device	

Three notes

	Note	Which consideration explains it?
One thing that reads		
One thing that confuses		
One thing to cut		

Observation checklist

- ☐ Found the landmark within 10 seconds
- ☐ Discovered at least one interaction without help
- ☐ Got lost or stuck (where?) _____
- ☐ Comfortable: no motion discomfort, text readable
- ☐ Could say what the world was about afterwards

Our next change